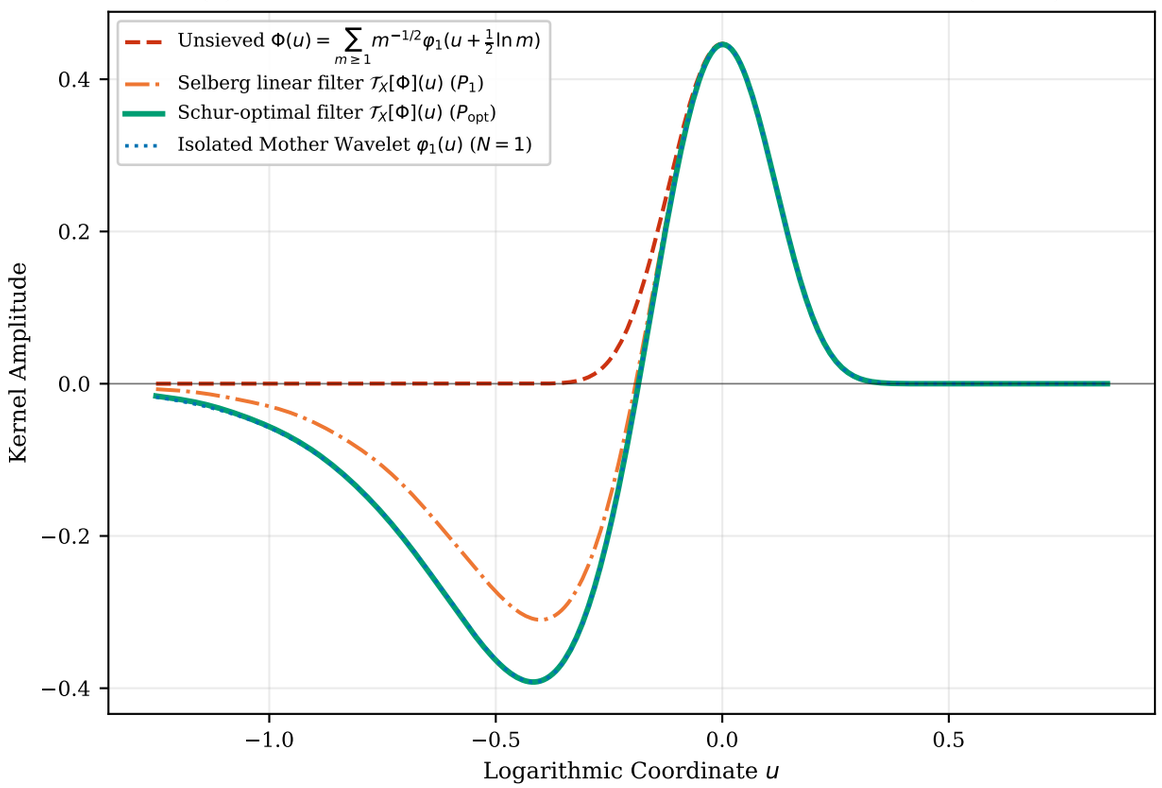
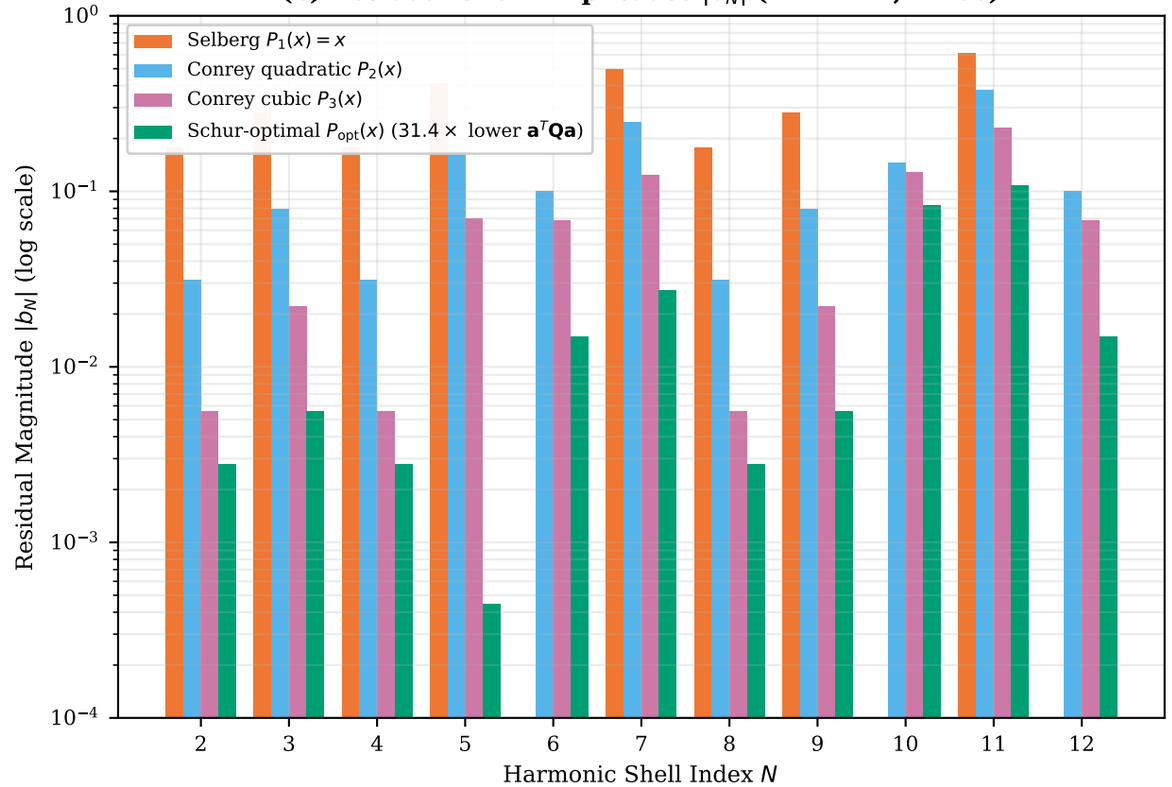
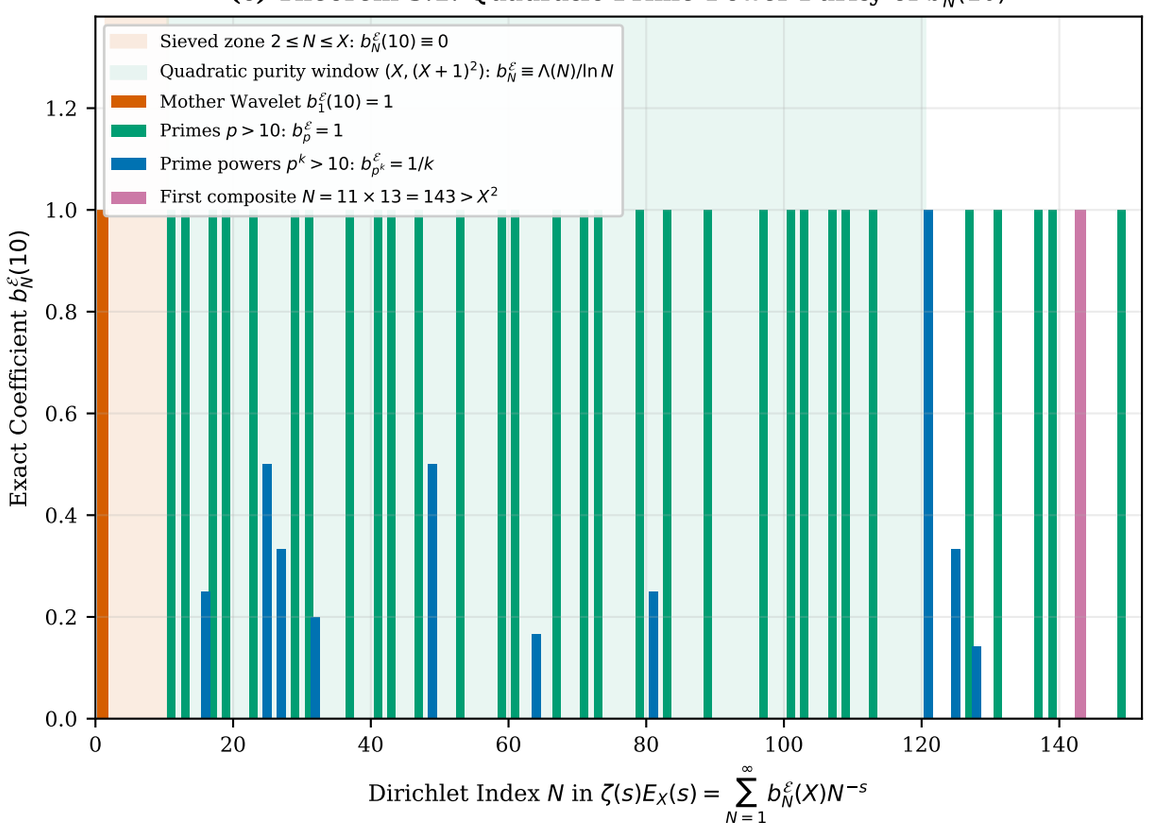
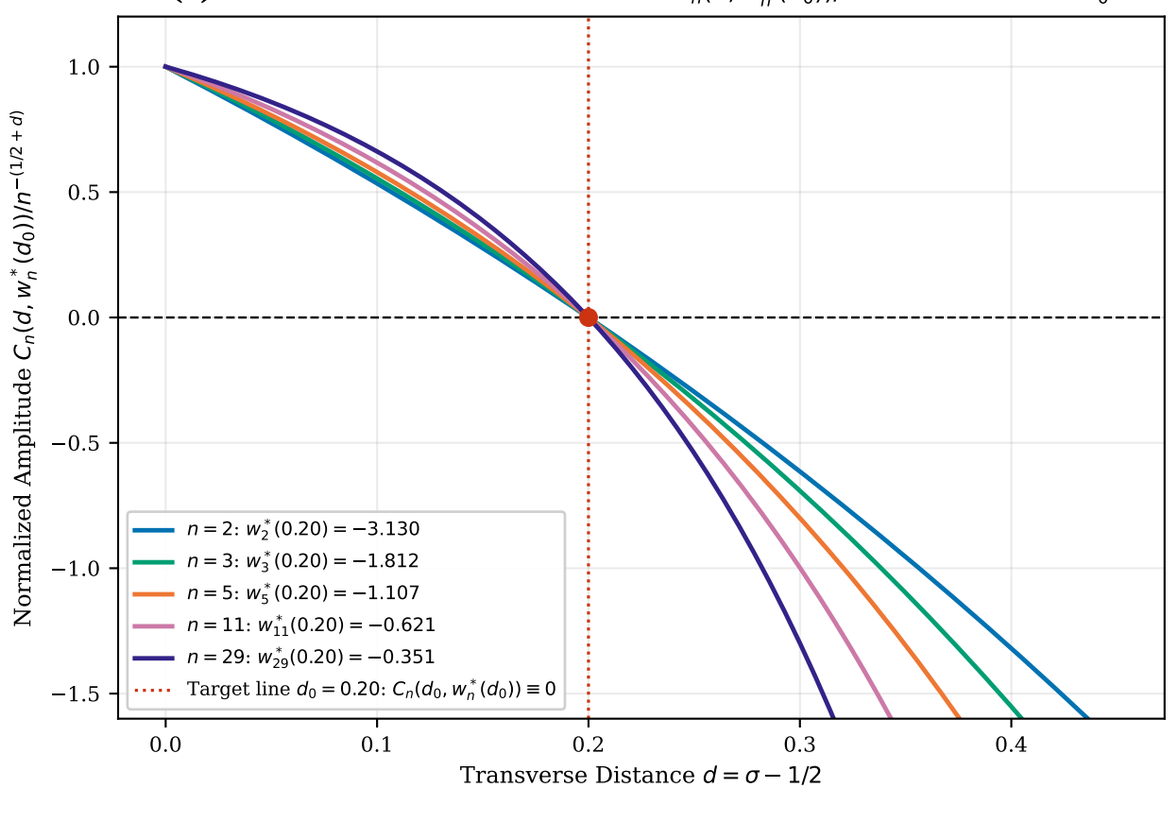

(a) Mother-Wavelet Shell Annihilation in the u-Domain ($X = 50$)
Unsieved $\Phi(u) = \sum_{m \geq 1} m^{-1/2} \varphi_1(u + \frac{1}{2} \ln m)$
Selberg linear filter $\mathcal{T}_X[\Phi](u)$ (P_1)
Schur-optimal filter $\mathcal{T}_X[\Phi](u)$ (P_{opt})
Isolated Mother Wavelet $\varphi_1(u)$ ($N = 1$)
Kernel Amplitude
Logarithmic Coordinate u


(b) Residual Shell Amplitudes $|b_N|$ ($2 \leq N \leq 12$, $X = 50$)
Selberg $P_1(x) = x$
Conrey quadratic $P_2(x)$
Conrey cubic $P_3(x)$
Schur-optimal $P_{\text{opt}}(x)$ (31.4× lower $\mathbf{a}^T\mathbf{Q}\mathbf{a}$)
Residual Magnitude $|b_N|$ (log scale)
Harmonic Shell Index N


(c) Theorem 3.1: Quadratic Prime-Power Purity of $b_N^{\mathcal{E}}(10)$
Sieved zone $2 \leq N \leq X$: $b_N^{\mathcal{E}}(10) \equiv 0$
Quadratic purity window $(X, (X+1)^2)$: $b_N^{\mathcal{E}} \equiv \Lambda(N)/\ln N$
Mother Wavelet $b_1^{\mathcal{E}}(10) = 1$
Primes $p > 10$: $b_p^{\mathcal{E}} = 1$
Prime powers $p^k > 10$: $b_{p^k}^{\mathcal{E}} = 1/k$
First composite $N = 11 \times 13 = 143 > X^2$
Exact Coefficient $b_N^{\mathcal{E}}(10)$
Dirichlet Index N in $\zeta(s)E_X(s) = \sum_{N=1}^{\infty} b_N^{\mathcal{E}}(X) N^{-s}$


(d) Theorem 3.2: Exact Coalescence $C_n(d, w_n^*(d_0))/n^{-(1/2+d)} = 0$ at $d = d_0$
Normalized Amplitude $C_n(d, w_n^*(d_0))/n^{-(1/2+d)}$
$n = 2$: $w_2^*(0.20) = -3.130$
$n = 3$: $w_3^*(0.20) = -1.812$
$n = 5$: $w_5^*(0.20) = -1.107$
$n = 11$: $w_{11}^*(0.20) = -0.621$
$n = 29$: $w_{29}^*(0.20) = -0.351$
Target line $d_0 = 0.20$: $C_n(d_0, w_n^*(d_0)) \equiv 0$
Transverse Distance $d = \sigma - 1/2$